

Table 1: ICI on the held-out EHRShot test set across the sparsity grid. Percentages, mean $\pm$ sample standard deviation over seeds 11/22/33. The grid is the same one used for MIMIC-III and MIMIC-IV.

K_1	K_2	AUROC (%) $\uparrow$				AUPRC (%) $\uparrow$				DCorr (%) $\uparrow$				MI (%) $\uparrow$			
		MORT	30M	LOS	READM	MORT	30M	LOS	READM	MORT	30M	LOS	READM	MORT	30M	LOS	READM
50	0	94.9 $\pm$ 0.4	77.6 $\pm$ 0.9	82.8 $\pm$ 0.3	32.8 $\pm$ 0.3	33.2 $\pm$ 4.2	18.5 $\pm$ 1.6	53.2 $\pm$ 0.4	32.8 $\pm$ 0.1	30.3 $\pm$ 1.1	23.2 $\pm$ 1.3	48.5 $\pm$ 0.2	26.8 $\pm$ 0.5	3.0 $\pm$ 0.2	2.0 $\pm$ 0.2	11.4 $\pm$ 0.3	5.4 $\pm$ 0.3
50	25	93.4 $\pm$ 0.6	75.4 $\pm$ 0.4	83.3 $\pm$ 0.8	32.9 $\pm$ 0.5	27.4 $\pm$ 2.2	15.4 $\pm$ 0.6	53.9 $\pm$ 1.7	32.6 $\pm$ 0.2	27.7 $\pm$ 1.0	20.2 $\pm$ 0.6	48.6 $\pm$ 1.0	27.1 $\pm$ 0.5	2.8 $\pm$ 0.2	1.5 $\pm$ 0.1	11.6 $\pm$ 0.6	5.6 $\pm$ 0.2
50	50	93.9 $\pm$ 0.6	76.5 $\pm$ 0.8	83.1 $\pm$ 0.4	32.4 $\pm$ 0.7	28.7 $\pm$ 3.1	17.7 $\pm$ 1.1	52.7 $\pm$ 0.7	32.5 $\pm$ 0.3	28.8 $\pm$ 1.1	22.1 $\pm$ 0.7	48.0 $\pm$ 0.1	27.9 $\pm$ 1.3	3.0 $\pm$ 0.2	2.0 $\pm$ 0.3	11.3 $\pm$ 0.3	5.7 $\pm$ 0.3
100	0	96.6 $\pm$ 0.4	78.9 $\pm$ 1.4	84.3 $\pm$ 0.2	32.0 $\pm$ 0.3	42.0 $\pm$ 0.3	23.2 $\pm$ 0.5	54.7 $\pm$ 1.2	32.1 $\pm$ 0.5	32.9 $\pm$ 0.6	24.8 $\pm$ 0.4	49.8 $\pm$ 0.5	27.1 $\pm$ 0.1	3.5 $\pm$ 0.2	2.2 $\pm$ 0.1	12.4 $\pm$ 0.5	5.9 $\pm$ 0.7
100	25	95.6 $\pm$ 0.4	77.4 $\pm$ 1.0	84.4 $\pm$ 0.5	31.5 $\pm$ 0.3	34.3 $\pm$ 3.2	19.4 $\pm$ 2.9	55.4 $\pm$ 1.0	31.9 $\pm$ 0.1	30.3 $\pm$ 1.2	22.7 $\pm$ 1.1	49.7 $\pm$ 0.7	27.9 $\pm$ 0.2	3.1 $\pm$ 0.1	2.0 $\pm$ 0.2	12.4 $\pm$ 0.3	6.4 $\pm$ 1.3
100	50	95.2 $\pm$ 0.4	78.4 $\pm$ 0.2	84.4 $\pm$ 0.6	31.5 $\pm$ 0.3	37.8 $\pm$ 7.8	21.1 $\pm$ 4.7	55.2 $\pm$ 1.6	31.8 $\pm$ 0.2	30.9 $\pm$ 0.7	23.4 $\pm$ 0.9	49.7 $\pm$ 1.0	28.4 $\pm$ 0.5	3.3 $\pm$ 0.2	2.2 $\pm$ 0.3	12.3 $\pm$ 0.3	6.7 $\pm$ 0.1
100	100	95.9 $\pm$ 0.6	77.9 $\pm$ 0.9	84.3 $\pm$ 0.4	30.8 $\pm$ 0.3	36.3 $\pm$ 3.8	22.0 $\pm$ 2.7	54.5 $\pm$ 0.6	31.7 $\pm$ 0.2	31.4 $\pm$ 0.8	23.8 $\pm$ 1.1	49.2 $\pm$ 0.7	29.2 $\pm$ 0.3	3.3 $\pm$ 0.2	2.1 $\pm$ 0.3	12.3 $\pm$ 0.5	6.9 $\pm$ 0.2
250	0	97.6 $\pm$ 0.1	80.2 $\pm$ 0.6	83.8 $\pm$ 0.6	32.3 $\pm$ 0.3	41.3 $\pm$ 1.3	23.4 $\pm$ 0.9	54.7 $\pm$ 1.5	31.1 $\pm$ 0.5	34.9 $\pm$ 0.5	26.3 $\pm$ 0.2	49.1 $\pm$ 1.3	27.2 $\pm$ 0.3	3.8 $\pm$ 0.1	2.4 $\pm$ 0.2	11.9 $\pm$ 0.3	6.8 $\pm$ 0.7
250	25	96.6 $\pm$ 0.8	78.5 $\pm$ 0.8	83.9 $\pm$ 0.5	30.2 $\pm$ 0.6	37.3 $\pm$ 3.6	20.8 $\pm$ 1.2	55.3 $\pm$ 1.3	30.4 $\pm$ 0.4	32.2 $\pm$ 1.3	23.8 $\pm$ 0.9	49.1 $\pm$ 0.6	29.8 $\pm$ 0.7	3.4 $\pm$ 0.2	2.1 $\pm$ 0.2	11.9 $\pm$ 0.3	8.6 $\pm$ 0.4
250	50	97.2 $\pm$ 0.2	78.9 $\pm$ 0.2	83.9 $\pm$ 0.5	30.2 $\pm$ 0.4	38.9 $\pm$ 1.9	21.8 $\pm$ 1.2	56.1 $\pm$ 0.2	30.1 $\pm$ 0.2	32.8 $\pm$ 1.3	24.3 $\pm$ 1.0	49.5 $\pm$ 0.2	29.5 $\pm$ 0.7	3.6 $\pm$ 0.1	2.3 $\pm$ 0.2	12.1 $\pm$ 0.2	8.6 $\pm$ 0.6
250	100	97.2 $\pm$ 0.4	78.8 $\pm$ 0.9	84.1 $\pm$ 0.7	30.0 $\pm$ 0.5	37.9 $\pm$ 2.8	21.6 $\pm$ 1.4	55.3 $\pm$ 1.2	30.0 $\pm$ 0.3	33.5 $\pm$ 0.9	24.8 $\pm$ 0.6	49.2 $\pm$ 1.1	30.1 $\pm$ 0.9	3.6 $\pm$ 0.2	2.4 $\pm$ 0.1	11.8 $\pm$ 0.4	8.1 $\pm$ 0.6
250	150	96.3 $\pm$ 0.8	78.5 $\pm$ 0.3	84.5 $\pm$ 1.1	30.1 $\pm$ 0.4	38.0 $\pm$ 4.3	21.7 $\pm$ 1.7	56.3 $\pm$ 1.7	30.2 $\pm$ 0.2	31.9 $\pm$ 1.9	23.7 $\pm$ 1.2	49.8 $\pm$ 1.7	29.4 $\pm$ 0.6	3.3 $\pm$ 0.4	2.0 $\pm$ 0.4	12.2 $\pm$ 0.9	7.4 $\pm$ 0.2
250	200	96.7 $\pm$ 1.0	79.0 $\pm$ 0.5	83.9 $\pm$ 0.9	30.1 $\pm$ 0.5	37.2 $\pm$ 5.7	22.0 $\pm$ 2.0	54.7 $\pm$ 2.3	30.1 $\pm$ 0.4	32.9 $\pm$ 1.9	24.6 $\pm$ 0.9	49.0 $\pm$ 1.9	29.4 $\pm$ 0.6	3.6 $\pm$ 0.6	2.2 $\pm$ 0.5	12.0 $\pm$ 1.1	7.9 $\pm$ 0.7
250	250	96.7 $\pm$ 0.5	78.3 $\pm$ 0.4	84.1 $\pm$ 0.6	30.3 $\pm$ 0.8	38.8 $\pm$ 1.7	21.7 $\pm$ 1.1	54.8 $\pm$ 1.0	30.4 $\pm$ 0.6	33.0 $\pm$ 0.4	24.1 $\pm$ 0.3	48.9 $\pm$ 0.8	29.7 $\pm$ 1.1	3.6 $\pm$ 0.2	2.3 $\pm$ 0.2	12.0 $\pm$ 0.6	8.2 $\pm$ 0.3
500	0	97.2 $\pm$ 0.3	79.0 $\pm$ 0.3	84.2 $\pm$ 1.3	31.2 $\pm$ 0.1	41.5 $\pm$ 3.5	22.5 $\pm$ 2.1	56.0 $\pm$ 2.9	29.9 $\pm$ 0.1	33.4 $\pm$ 1.1	24.8 $\pm$ 0.9	50.0 $\pm$ 1.9	28.8 $\pm$ 0.5	3.7 $\pm$ 0.1	2.2 $\pm$ 0.0	12.4 $\pm$ 0.7	8.0 $\pm$ 1.0
500	25	97.1 $\pm$ 0.3	78.2 $\pm$ 0.3	84.0 $\pm$ 1.2	30.0 $\pm$ 0.5	39.0 $\pm$ 3.8	21.8 $\pm$ 1.5	55.9 $\pm$ 0.6	29.5 $\pm$ 0.1	32.4 $\pm$ 1.3	24.0 $\pm$ 0.9	49.6 $\pm$ 0.8	30.1 $\pm$ 0.3	3.7 $\pm$ 0.1	2.4 $\pm$ 0.2	12.5 $\pm$ 0.4	8.6 $\pm$ 0.5
500	50	97.0 $\pm$ 0.7	77.7 $\pm$ 0.4	83.6 $\pm$ 0.9	30.4 $\pm$ 0.2	38.0 $\pm$ 4.6	21.0 $\pm$ 2.4	55.3 $\pm$ 1.3	29.4 $\pm$ 0.1	32.6 $\pm$ 1.7	24.0 $\pm$ 1.1	49.1 $\pm$ 1.2	29.6 $\pm$ 0.5	3.7 $\pm$ 0.2	2.1 $\pm$ 0.3	11.7 $\pm$ 0.8	8.2 $\pm$ 0.3
500	100	97.4 $\pm$ 0.1	78.4 $\pm$ 0.4	83.2 $\pm$ 1.6	30.2 $\pm$ 0.7	40.0 $\pm$ 1.2	21.7 $\pm$ 1.1	54.5 $\pm$ 2.8	29.6 $\pm$ 0.2	32.9 $\pm$ 0.0	24.2 $\pm$ 0.5	48.3 $\pm$ 2.0	29.9 $\pm$ 0.8	3.7 $\pm$ 0.1	2.1 $\pm$ 0.1	11.5 $\pm$ 0.9	8.2 $\pm$ 1.0
500	150	97.1 $\pm$ 0.6	77.8 $\pm$ 0.9	82.5 $\pm$ 0.4	30.3 $\pm$ 0.1	41.1 $\pm$ 0.1	22.5 $\pm$ 0.4	53.0 $\pm$ 1.3	29.5 $\pm$ 0.3	33.4 $\pm$ 1.4	24.4 $\pm$ 0.9	47.1 $\pm$ 0.4	29.9 $\pm$ 0.4	3.7 $\pm$ 0.3	2.2 $\pm$ 0.2	10.9 $\pm$ 0.2	9.0 $\pm$ 0.5
500	200	97.2 $\pm$ 0.4	78.1 $\pm$ 0.6	83.4 $\pm$ 0.9	30.6 $\pm$ 0.6	39.3 $\pm$ 4.0	21.7 $\pm$ 2.1	54.1 $\pm$ 1.3	29.6 $\pm$ 0.3	33.1 $\pm$ 1.5	24.2 $\pm$ 1.0	48.0 $\pm$ 1.0	29.3 $\pm$ 1.2	3.6 $\pm$ 0.3	2.1 $\pm$ 0.3	11.3 $\pm$ 0.5	8.3 $\pm$ 0.5
500	250	97.0 $\pm$ 0.6	77.2 $\pm$ 1.1	82.5 $\pm$ 0.2	30.1 $\pm$ 0.1	39.3 $\pm$ 5.3	21.7 $\pm$ 2.9	53.4 $\pm$ 0.7	29.4 $\pm$ 0.1	33.0 $\pm$ 1.2	23.9 $\pm$ 0.8	47.3 $\pm$ 0.5	30.4 $\pm$ 0.1	3.5 $\pm$ 0.3	2.1 $\pm$ 0.1	11.0 $\pm$ 0.2	8.3 $\pm$ 0.4
500	300	96.7 $\pm$ 0.8	77.3 $\pm$ 0.9	83.4 $\pm$ 1.2	30.2 $\pm$ 0.3	37.0 $\pm$ 2.3	20.7 $\pm$ 1.2	54.7 $\pm$ 1.7	29.5 $\pm$ 0.3	33.1 $\pm$ 1.8	24.2 $\pm$ 1.0	48.4 $\pm$ 1.7	29.5 $\pm$ 0.1	3.4 $\pm$ 0.3	2.0 $\pm$ 0.2	11.9 $\pm$ 0.6	8.4 $\pm$ 0.7
500	350	96.7 $\pm$ 0.6	77.4 $\pm$ 0.4	83.0 $\pm$ 1.3	30.2 $\pm$ 0.6	38.9 $\pm$ 5.1	22.1 $\pm$ 2.2	53.4 $\pm$ 2.3	29.7 $\pm$ 0.3	33.2 $\pm$ 1.9	24.5 $\pm$ 1.3	47.5 $\pm$ 2.2	29.8 $\pm$ 0.7	3.5 $\pm$ 0.3	2.1 $\pm$ 0.3	11.8 $\pm$ 1.3	8.6 $\pm$ 0.3
500	400	96.7 $\pm$ 0.7	77.5 $\pm$ 0.4	82.0 $\pm$ 2.1	30.6 $\pm$ 0.5	41.7 $\pm$ 6.3	23.4 $\pm$ 2.9	52.8 $\pm$ 3.4	29.6 $\pm$ 0.1	33.6 $\pm$ 1.3	24.6 $\pm$ 0.7	46.7 $\pm$ 3.1	28.9 $\pm$ 0.4	3.6 $\pm$ 0.2	2.3 $\pm$ 0.3	11.1 $\pm$ 1.1	8.0 $\pm$ 0.5
500	450	96.7 $\pm$ 0.8	77.6 $\pm$ 0.9	82.7 $\pm$ 1.7	29.6 $\pm$ 0.9	38.1 $\pm$ 1.5	20.7 $\pm$ 0.6	53.1 $\pm$ 2.4	29.4 $\pm$ 0.1	32.9 $\pm$ 0.5	23.7 $\pm$ 0.3	46.9 $\pm$ 2.1	30.3 $\pm$ 0.9	3.5 $\pm$ 0.1	1.9 $\pm$ 0.1	11.5 $\pm$ 1.0	8.4 $\pm$ 0.2
500	500	96.5 $\pm$ 0.6	77.0 $\pm$ 0.6	83.2 $\pm$ 1.1	29.3 $\pm$ 0.8	36.2 $\pm$ 1.2	19.7 $\pm$ 1.3	54.2 $\pm$ 2.4	29.3 $\pm$ 0.2	32.4 $\pm$ 0.3	23.3 $\pm$ 0.4	47.9 $\pm$ 1.9	30.6 $\pm$ 1.1	3.4 $\pm$ 0.1	2.0 $\pm$ 0.2	11.5 $\pm$ 0.3	8.7 $\pm$ 0.5
1000	0	97.1 $\pm$ 0.2	78.0 $\pm$ 0.3	84.1 $\pm$ 2.7	30.7 $\pm$ 1.5	38.0 $\pm$ 2.0	21.0 $\pm$ 1.0	56.1 $\pm$ 4.1	29.2 $\pm$ 0.5	32.2 $\pm$ 1.0	24.2 $\pm$ 0.6	49.6 $\pm$ 3.5	29.1 $\pm$ 1.5	3.7 $\pm$ 0.0	2.2 $\pm$ 0.1	12.5 $\pm$ 1.8	8.6 $\pm$ 0.4
1000	25	96.9 $\pm$ 0.1	77.1 $\pm$ 0.2	84.4 $\pm$ 1.1	29.8 $\pm$ 0.7	38.1 $\pm$ 1.3	20.5 $\pm$ 0.5	56.1 $\pm$ 1.9	28.8 $\pm$ 0.2	31.5 $\pm$ 0.4	22.9 $\pm$ 0.3	49.4 $\pm$ 1.8	30.6 $\pm$ 0.8	3.5 $\pm$ 0.0	2.2 $\pm$ 0.2	12.5 $\pm$ 1.5	8.1 $\pm$ 1.5
1000	50	96.6 $\pm$ 0.3	77.4 $\pm$ 0.6	85.7 $\pm$ 0.6	29.3 $\pm$ 0.5	35.2 $\pm$ 3.7	18.9 $\pm$ 0.9	58.9 $\pm$ 1.0	28.7 $\pm$ 0.1	30.7 $\pm$ 1.1	22.5 $\pm$ 0.4	51.5 $\pm$ 0.9	30.9 $\pm$ 0.7	3.4 $\pm$ 0.1	1.8 $\pm$ 0.1	13.0 $\pm$ 0.7	8.6 $\pm$ 0.4
1000	100	96.9 $\pm$ 0.3	77.7 $\pm$ 0.7	85.3 $\pm$ 0.9	29.7 $\pm$ 0.9	37.6 $\pm$ 3.2	19.9 $\pm$ 0.8	58.0 $\pm$ 2.4	28.9 $\pm$ 0.3	31.4 $\pm$ 1.2	23.1 $\pm$ 0.7	50.8 $\pm$ 1.7	30.6 $\pm$ 0.9	3.5 $\pm$ 0.2	2.3 $\pm$ 0.1	12.6 $\pm$ 0.7	8.5 $\pm$ 0.4
1000	150	97.0 $\pm$ 0.2	77.1 $\pm$ 0.9	82.3 $\pm$ 1.5	31.0 $\pm$ 1.1	37.1 $\pm$ 2.4	21.1 $\pm$ 1.4	53.6 $\pm$ 1.6	29.2 $\pm$ 0.3	32.5 $\pm$ 1.0	23.4 $\pm$ 1.0	47.5 $\pm$ 1.4	29.1 $\pm$ 0.9	3.6 $\pm$ 0.2	2.1 $\pm$ 0.2	11.3 $\pm$ 0.7	8.2 $\pm$ 0.3
1000	200	96.8 $\pm$ 0.3	77.5 $\pm$ 0.3	84.6 $\pm$ 0.5	29.9 $\pm$ 0.6	37.0 $\pm$ 3.4	19.7 $\pm$ 1.1	56.1 $\pm$ 1.1	29.0 $\pm$ 0.3	32.0 $\pm$ 1.1	23.3 $\pm$ 0.8	49.8 $\pm$ 1.0	30.1 $\pm$ 0.7	3.4 $\pm$ 0.1	2.2 $\pm$ 0.4	12.1 $\pm$ 0.4	8.5 $\pm$ 0.8
1000	250	96.9 $\pm$ 0.5	77.8 $\pm$ 1.1	84.9 $\pm$ 1.1	29.2 $\pm$ 0.8	36.6 $\pm$ 5.1	19.3 $\pm$ 2.1	57.0 $\pm$ 3.1	28.8 $\pm$ 0.3	31.6 $\pm$ 1.8	23.2 $\pm$ 1.4	50.1 $\pm$ 2.0	31.0 $\pm$ 0.9	3.4 $\pm$ 0.2	2.0 $\pm$ 0.2	12.4 $\pm$ 1.1	8.8 $\pm$ 0.9
1000	300	96.6 $\pm$ 0.4	77.7 $\pm$ 0.5	85.1 $\pm$ 1.0	29.3 $\pm$ 0.5	35.8 $\pm$ 4.2	19.0 $\pm$ 1.6	57.8 $\pm$ 2.8	28.7 $\pm$ 0.1	31.5 $\pm$ 1.5	23.0 $\pm$ 0.8	50.5 $\pm$ 2.1	31.2 $\pm$ 0.5	3.4 $\pm$ 0.2	2.1 $\pm$ 0.1	12.5 $\pm$ 0.4	8.6 $\pm$ 0.4
1000	350	96.8 $\pm$ 0.5	77.5 $\pm$ 0.2	84.7 $\pm$ 0.9	29.4 $\pm$ 0.4	38.1 $\pm$ 2.0	20.5 $\pm$ 0.5	56.7 $\pm$ 1.4	28.7 $\pm$ 0.1	32.2 $\pm$ 1.1	23.5 $\pm$ 0.4	50.0 $\pm$ 1.8	30.8 $\pm$ 0.2	3.5 $\pm$ 0.1	2.1 $\pm$ 0.0	12.1 $\pm$ 0.9	8.4 $\pm$ 0.1
1000	400	96.7 $\pm$ 0.4	77.4 $\pm$ 0.3	8													